Training-Free Token Reduction on MotionBench

Results

HuVLLM Project

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Eleven training-free video-LLM token-reduction methods, evaluated on MotionBench (8,052 questions, 4,018 scoreable) across two backbones — LLaVA-OV-7B and Qwen3-VL-8B — at 32 frames. Results only: full methodology, the verification gate, and limitations are in `paper.tex`. At $n = 4{,}018$ the binomial 95% confidence half-width is ±1.54 points; * marks a difference significant by McNemar's test on paired predictions ($\chi^2 \geq 3.84$, $p < 0.05$).

Contents

1 Common settings

Held fixed across every method and both backbones:

- **32 frames**, sampled uniformly per video.
- **Greedy decoding** (`do_sample=False`, `max_new_tokens=16`), batch size 1.
- **Letter-match scoring** (A–D) against ground truth; the 4,034 `NA` items are excluded, leaving 4,018 scoreable.
- Methods exposing a single retention knob are standardized to **15% nominal retention**; methods with no such knob (frame-selection methods, or methods whose budget is fixed internally) keep their published defaults — forcing 15% on those would break paper fidelity.

Per-method settings, mechanisms and measured token budgets are in Table 6 (§4).

2 Master table

Table 1 and Table 2: every method at its standardized 15% or published default, one table per backbone. § AIM's released code ships two active merge steps, so its LLaVA-OV cell runs at 25%. MDP3 and VideoITG select frames, not tokens, so they have no retention percentage. Full retention sweeps are in §2.1.

Table 1: LLaVA-OV-7B, 32 frames, 15% nominal retention except where noted. AO = Action Order, CM = Camera Motion, LM = Location-related Motion, MR = Motion Recognition, MO = Motion-related Objects, RC = Repetition Count (% correct within category, chance = 25%). Δ is vs. baseline. * = significant. § AIM runs at 25% retention (two active merge steps ship in the released code). † PruneVID-OV runs at `cluster_ratio=0.5` (50% retention), the authors' only published point, never 15%.

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
Baseline	*52.66*	—	*40.5*	*45.2*	*55.5*	*57.0*	*71.2*	*23.8*
DyCoke	53.36	+0.70	38.9	48.8	55.7	58.2	70.9	25.3
FlashVID	53.31	+0.65	39.9	45.7	53.8	58.0	71.7	28.2
HoliTom	53.14	+0.47	40.8	49.6	52.6	57.0	71.4	27.3
MDP3	53.06	+0.40	40.5	49.6	53.5	56.8	71.6	26.5
VideoITG	52.86	+0.20	40.1	47.0	53.7	57.6	70.1	26.8
AIM§	52.86	+0.20	41.4	48.1	54.2	57.0	71.9	22.3
STTM	51.72	–0.95	39.9	48.1	53.8	53.5	70.6	28.8
PruneVID-OV*†	38.20	–14.46	32.0	34.8	35.5	38.4	52.9	27.3
FastV*	36.78	–15.88	32.8	31.9	34.1	35.7	53.8	25.3

Table 2: Qwen3-VL-8B, 32 frames, 15% nominal retention except where noted. All ten portable methods. VisionZip is the contextual half only — Qwen3-VL's vision tower has no CLS token, so the dominant half is not implementable; not the published method. FlashVID and STTM run the authors' own code at published settings (§4). † PruneVID runs at `cluster_ratio=0.5` (50% retention), the authors' only published point, never 15%.

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
Baseline	*62.52*	—	*46.1*	*63.1*	*65.0*	*67.3*	*79.0*	*33.8*
PruneVID†	62.17	–0.35	46.1	62.9	64.1	67.1	79.1	32.5
DyCoke	61.85	–0.67	45.1	62.6	63.9	66.4	78.4	34.5
STTM*	61.60	–0.92	45.9	61.6	64.8	66.0	77.1	34.8
HoliTom*	60.33	–2.19	44.7	61.3	61.0	66.4	76.2	29.0
MDP3*	59.66	–2.86	44.9	64.7	62.8	64.0	77.4	23.0
FlashVID*	59.36	–3.16	43.5	57.4	62.6	65.6	77.2	23.5
FastV*	59.01	–3.51	46.2	61.0	61.9	63.5	72.2	30.5
VisionZip (ctx.)*	58.81	–3.71	43.2	57.4	61.9	64.2	74.3	29.5
VideoITG*	56.35	–6.17	45.3	53.5	59.3	60.2	75.4	22.2
AIM*	55.97	–6.55	45.9	56.6	59.5	58.3	72.2	27.0

2.1 Retention sweep

Three methods (FastV, FlashVID, HoliTom) have a retention knob the authors validate at more than one point; each is run at every point they publish, not just 15%. Table 3 and Table 4 are that data, one table per backbone; baseline is in Table 1/ 2 above, not repeated here.

Table 3: LLaVA-OV-7B retention sweep (baseline 52.66%). * = significant. FastV's range is its authors' full published boundary; the other two publish 10/15/20/25% on every backbone they support.

Method	10%	15%	20%	25%	50%	75%
FastV	36.06*	36.78*	—	36.73*	36.34*	35.69*
FlashVID	52.51	53.31	53.71*	53.36	—	—
HoliTom	52.76	53.14	53.06	53.41	—	—

Table 4: Qwen3-VL-8B retention sweep (baseline 62.52%). * = significant. 20% was never run on Qwen3-VL for any method. ‡ FlashVID's 10%/25% cells predate the authors'-code fix (§4): the incomplete reimplementation, not the verified method that produced the 15% cell — not directly comparable to it, despite the coincidental match at 25%.

Method	10%	15%	20%	25%
FastV	56.92*	59.01*	—	60.60*
FlashVID‡	54.73*	59.36*	—	59.36*
HoliTom	60.05*	60.33*	—	60.43*

FastV's 50%/75% rows are K=2/R=50% and K=2/R=75%, the authors' own most- published operating points; both land in the same collapse band as every other FastV cell. FlashVID's 20% LLaVA-OV cell is the one retention point in this report that crosses significance in the direction of a gain ($\chi^2 = 3.91$, 236 fixed vs. 194 broke); HoliTom's 20% does not ($\chi^2 = 0.43$). Both gate clean; it survives at only one of four points, so it is reported because it crosses the threshold, not because it changes the picture that FlashVID is accuracy-neutral on LLaVA-OV. FlashVID's two Qwen3-VL reimplementation cells (54.73%, 59.36%) predate the authors'-code fix; the 25% cell's match to the verified 15% cell is coincidence, not a duplicated run — the two share only 5,725 of 8,052 predictions.

PruneVID's authors publish exactly one operating point, `cluster_ratio=0.5` (its Table 1/2 row); Table 5 is what we additionally ran, for context, not a published sweep.

Table 5: PruneVID beyond its one published point (not authors'-validated). Sorted by the ratio that actually ran, Qwen3-VL is monotonic; an earlier revision of this report mislabeled the 50% column "15%" and flagged the resulting order as a non-monotonic anomaly, which was an artifact of that mislabeling, not a property of the method.

Backbone	10%	25%	50% (published)
LLaVA-OV-7B (OV port)	38.10*	38.38*	38.20*
Qwen3-VL-8B	60.23*	61.40*	62.17

Tripling FastV's budget from 10% to 25% moves it 3.68 points on Qwen3-VL, against no more than 0.85 points for any LLaVA-OV method over the same range — the two backbones respond to retention completely differently.

3 Figures

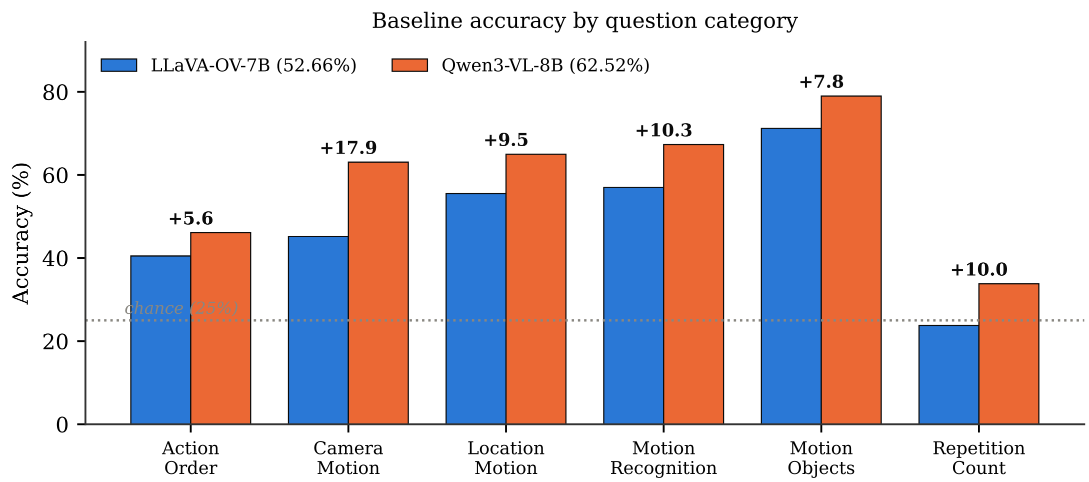

Figure 1: Baseline accuracy by category, both backbones (bar).

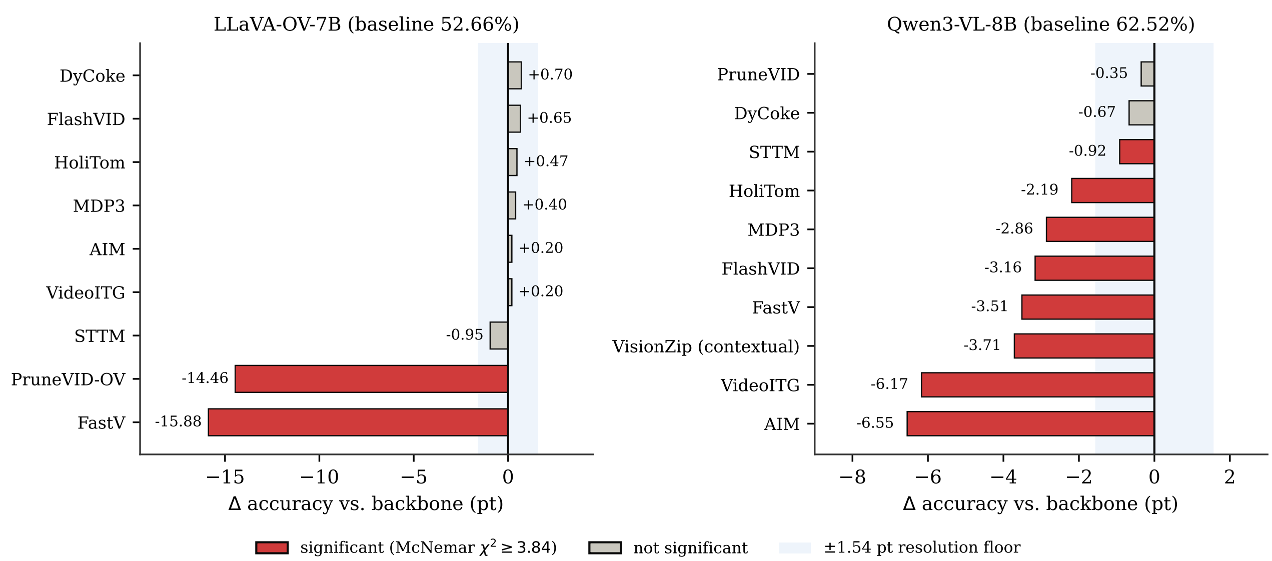

Figure 2: Change vs. backbone, every method, both backbones (bar). Red = significant by McNemar's test; the shaded band is the ±1.54-point resolution floor.

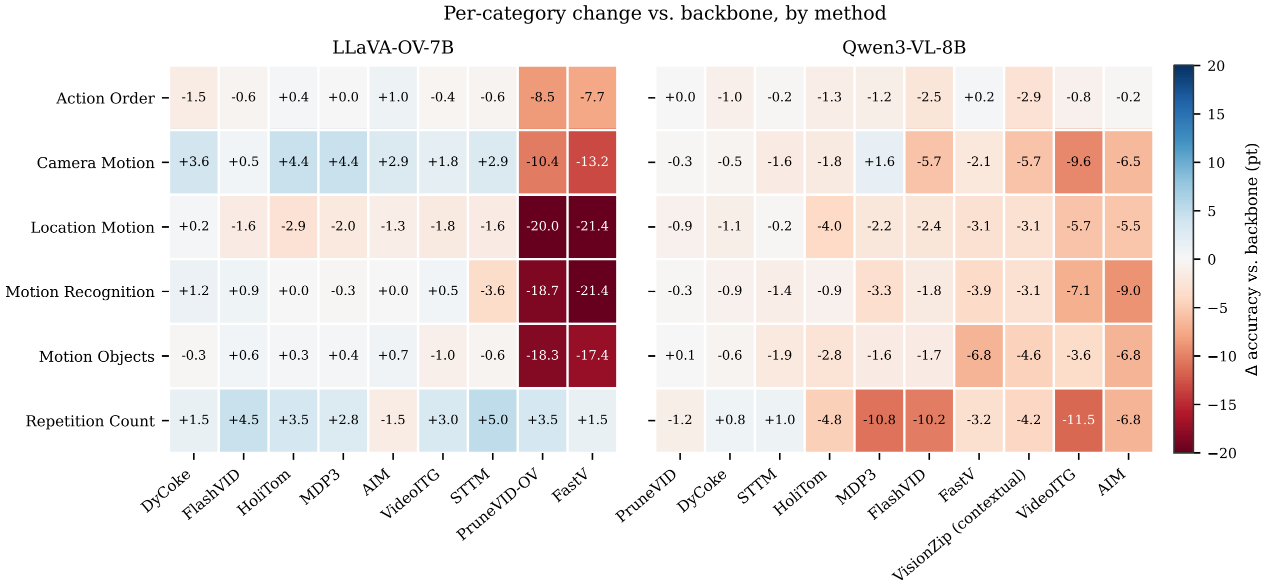

Figure 3: Per-category change vs. backbone, by method (heatmap). Blue = gain, red = loss.

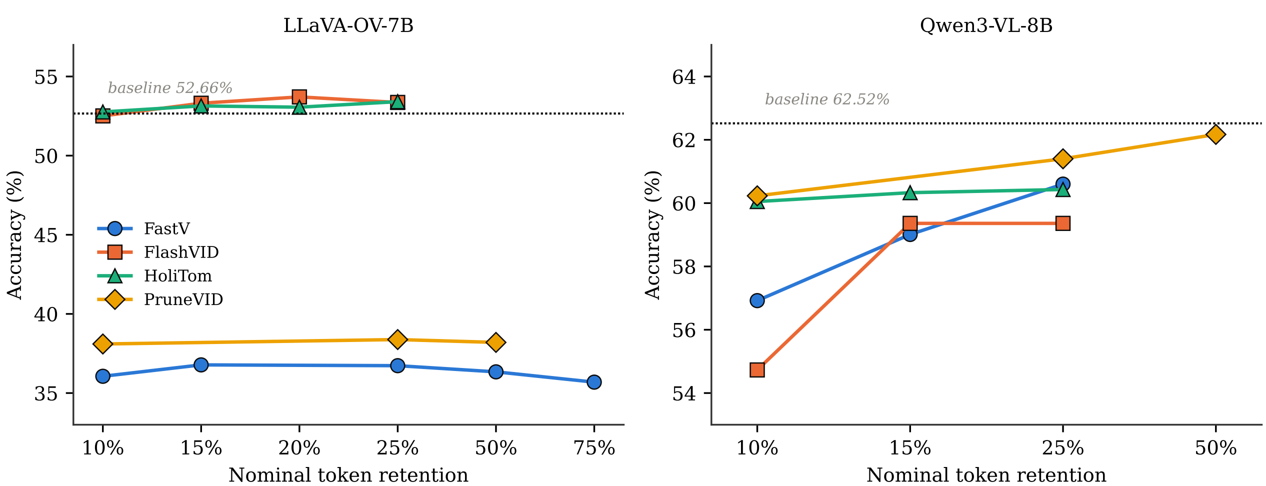

Figure 4: Accuracy against nominal token retention, the four methods with a retention knob (dot + line).

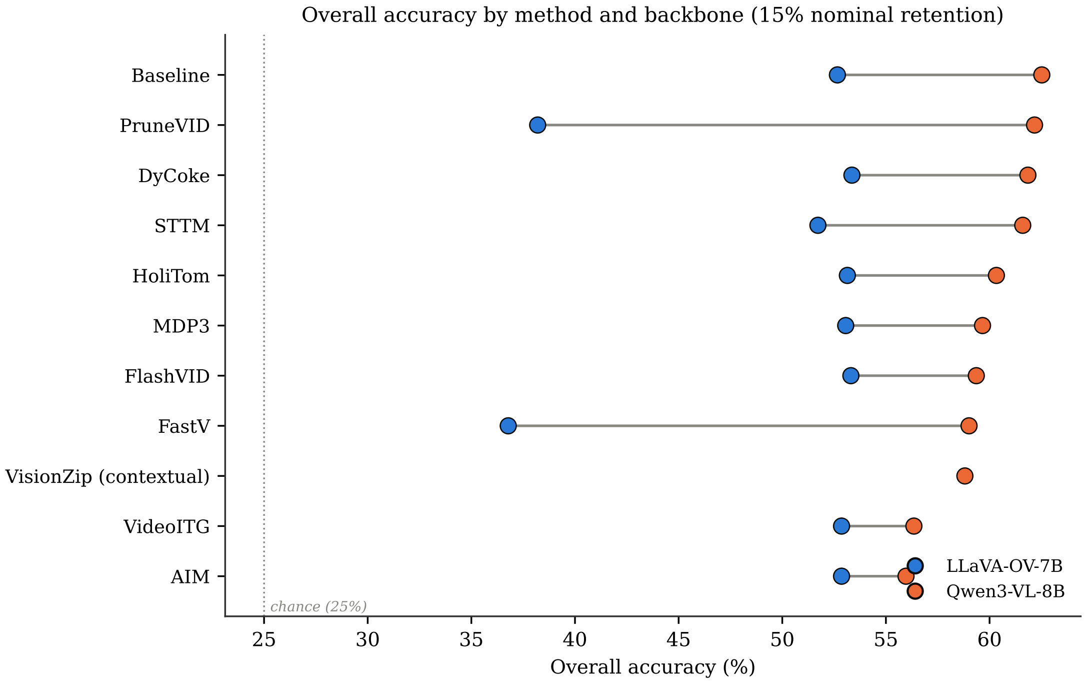

Figure 5: Overall accuracy by method and backbone at 15% nominal retention (Cleveland dot plot). Rows sorted by Qwen3-VL accuracy; the line spans each method's two backbone results.

4 Per-method settings

Table 6 is the mechanism and exact configuration behind every row above.

5 Correct/incorrect vs. baseline

Fixed = wrong under the baseline, correct under the method. *Broke* = correct under the baseline, wrong under the method. Both are McNemar discordant pairs on the 4,018 scoreable items. *Differ* counts predictions unequal to the baseline's over all 8,052 rows (0 would mean a silent no-op — see `paper.tex` for the verification gate this guards against).

Table 6: Per-method mechanism, key parameters (as run in this report), and measured visual-token budget. Budgets are read from each method's execution log on Qwen3-VL-8B (11,664 visual input tokens at 32 frames), not from its CLI flag — nominal 15% describes the configured knob, not necessarily what is actually retained. Frame-selection methods have no token budget to measure; they instead reduce the number of frames fed to the model.

Method	Mechanism	Key parameters (this report)	Measured budget
HoliTom	pre-LLM segment retain + in-LLM merge	`RETAIN_RATIO=0.15 T=0.80 HOLITOM_k=18 HOLITOM_r=0.5`	7.5% (875 tok.)
STTM	quadtree spatio-temporal merge	LLaVA-OV: `thresh=0.85 temporal=0.65 root=1`. Qwen3-VL: `thresh=0.85 temporal=0.65 root=2` — a published setting (`s3_sttm_pub_run`), replacing an earlier off-spec `root=0, temporal=--1.0` cell	9.0% (1,050 tok.) on LLaVA-OV; not re-measured at the corrected Qwen3-VL setting
FastV	in-LLM attention rerank, layer 2	`K=2, R=0.85` (keep 15%). K=2/R=50% (36.34%) and K=2/R=75% (35.69%) were also run and land in the same collapse band (Table 3)	15.0% (1,750 tok.)
FlashVID	pre-LLM temporal-segment merge + inner-LLM compression, layer 28	authors' `flashvid()` package: `retention_ratio=0.15 alpha=0.7 temporal_threshold=0.8 token_selection_method=attn_div pruning_layer=28 llm_retention_ratio=0.1 expansion=1.25 min_segment_num=4`	15.0% (1,750 tok.) pre-LLM; the layer-28 stage compresses surviving visual tokens by a further 90% mid-network, not captured by this single figure
AIM	bipartite merge + PageRank prune	2-step merge schedule compiled in; no CLI retention knob. Qwen3-VL 15%; LLaVA-OV 25% (ships two active steps)	15.0% (1,750 tok.)
VisionZip (contextual)	key-similarity merge in the vision tower	`dominant=54 contextual=10` at 8 frames (native LLaVA-1.5-7B form); Qwen3-VL runs the contextual half only — no CLS token to select the dominant half	15.0% (1,750 tok.)
DyCoke	temporal merge + KV prune, layer 3	`l=3 p=0.7 k=0.7`; no single retention knob, net ∼49%	49.0% (5,716 tok.)
PruneVID	DPC-KNN cluster merge, layer 10	`cluster=0.50 seg=0.25`; LLaVA-OV port has no upstream reference (upstream targets PLLaVA only)	50.0% (5,832 tok.)
MDP3	frame selection (DPP)	`pool=32 select=8` — selects frames, not tokens	n/a (8 of 32 frames)
VideoITG	grounded frame selection	512 sampled → 32 selected	n/a (32 of 512 sampled)
DyTo[†]	FINCH clustering + ToMe token merge	`temporal_aggregation=spatial_tome_finch_dynamic_all_frms`; own Vicuna-7B backbone	∼3,680 tok. from 100 input frames

[†] DyTo does not run on either standardized backbone; listed here for completeness (see Table 8).

Table 7: Paired-prediction statistics against each backbone's baseline.

Method	Differ	Fixed	Broke	χ^2	Significant
LLaVA-OV-7B					
DyCoke	1,031	170	142	2.34	no
FlashVID	1,610	268	242	1.23	no
HoliTom	1,838	299	280	0.56	no
MDP3	1,452	246	230	0.47	no
VideoITG	1,193	192	184	0.13	no
AIM	1,406	227	219	0.11	no
STTM	4,859	329	367	1.97	no
PruneVID-OV	4,536	473	1,054	220.30	yes
FastV	4,605	475	1,113	255.52	yes
Qwen3-VL-8B					
PruneVID	1,074	56	70	1.34	no
DyCoke	1,048	83	110	3.50	no
HoliTom	1,657	131	219	21.63	yes
MDP3	1,626	159	274	30.01	yes
FastV	2,029	149	290	44.65	yes
VisionZip (contextual)	2,035	191	340	41.25	yes
STTM	1,510	133	170	4.28	yes
FlashVID	2,069	157	284	36.00	yes
VideoITG	2,522	206	454	92.44	yes
AIM	2,861	194	457	105.44	yes
Backbone vs. backbone					
Qwen3-VL-8B vs. LLaVA-OV-7B	7,081	810	414	127.47	yes

Divergence volume does not predict damage. STTM on LLaVA-OV changes 4,859 predictions — more than FastV's 4,605 — yet loses 0.94 points and is not significant, because its changes fall almost evenly in both directions (329 fixed, 367 broken). FastV changes a comparable number destructively (475 fixed, 1,113 broken).

6 Methods on non-standard backbones

Table 8: Runs on backbones other than the two standardized ones. Not comparable to Tables 1–2; no Δ is given because no matched baseline exists for any row. PruneVID's released code targets PLLaVA only — this is the authors' real backbone and the one cell with an upstream reference to validate against, unlike the LLaVA-OV port (PruneVID-OV, Table 1).

Method	Backbone	Overall	AO	CM	LM	MR	MO	RC
VisionZip	LLaVA-1.5-7B	40.09	33.7	33.0	37.2	41.1	57.4	25.8
PruneVID	PLLaVA-7B	44.13	35.6	33.0	41.0	47.0	63.3	26.5
DyTo (recon. TW-FINCH)	Vicuna-7B	42.06	32.4	33.0	42.1	43.0	61.7	26.0

7 Notable

The backbone swap moves accuracy 9.86 points — more than any token-reduction method moves it, either direction, either backbone, any retention level tested. Rankings do not transfer: PruneVID is the worst LLaVA-OV result (38.20%) and the best Qwen3-VL one (−0.35); Figure 5 shows the crossing lines directly. FastV and PruneVID-OV collapse specifically on LLaVA-OV (37–38%, >14 points below baseline, significant), while every other LLaVA-OV method is statistically indistinguishable from the backbone; FastV's own published range does not save it (36.3–36.4% at 50% and 75% keep, Table 3). Qwen3-VL inverts the pattern: every method loses, eight of ten losses are significant, but none approaches the LLaVA-OV collapse — the largest Qwen3-VL loss (AIM, -6.55) is smaller than the smallest LLaVA-OV collapse.

Only three methods qualify as having an authors'-validated retention sweep (§2.1); the other five with any retention-like parameter do not. **PruneVID** publishes exactly one ratio. **DyCoke** has no single retention knob, only one fixed three-parameter recipe. **AIM** has no knob at all on LLaVA-OV and an unpublished, self-ported one on Qwen3-VL. **STTM** and **VisionZip** are configured per-benchmark or as fixed token budgets, not a sweepable percentage.